HAMMAM OMER

Medical Student | Medical AI Research & Systems Engineering

Omdurman, Sudan | Email: hammamomer304@gmail.com | GitHub: github.com/HammamNM | Explorer: hammamnm.github.io/reasoning-debugger
Cloud Platform: hammam16.pythonanywhere.com | Preprint DOI: 10.5281/zenodo.23835945 | Blog: nexoramed.online

Executive Profile: Third-year medical student and Python systems builder specializing in Clinical LLM Safety Auditing, Neuro-Symbolic Verification, and Automated Rubric Engineering. Architect of two live open-source tools (Deterministic-First Hybrid Clinical Auditor & Cloud Clinical Reasoning Platform), backed by a DOI-registered benchmark across 420 emergency clinical evaluations anchored to Class I international society guidelines.

71.7% (38/53)	0 / 1,767	100% / 100%	2 LIVE TOOLS
JUDGE BLINDNESS EXPOSED	VERBATIM GATE HALLUCINATIONS	FROZEN-RULE BENCHMARK SPEC/SENS	CLOUD APP + TRACE EXPLORER

INDEPENDENT RESEARCH & SYSTEMS ENGINEERING EXPERIENCE

1. Deterministic-First Hybrid Clinical Auditor (v0.2-frozen-rules) [Interactive 420-Trace Explorer] | [GitHub Repo] | [50-Scenario Specs]

Independent Lead Researcher & System Architect | Neuro-Symbolic Clinical Safety Auditing & Benchmark Engineering (2026)

- **Two-Stage Neuro-Symbolic Architecture:** Engineered a decoupled evaluation pipeline isolating constrained semantic extraction (Stage I) from pure-Python deterministic rule verification (Stage II, <2ms CPU latency) across 420 paired emergency management plans on 6 models (gpt-oss-120b/20b, gemini-3.5/3.1-flash-lite, gemma-4-26b, qwen3.8-27b).
- **Exposed LLM-as-a-Judge Blindness:** Demonstrated empirically on the held-out 35-scenario test split that standalone LLM judges missed 38 of 53 criterion-unsafe responses (71.7% false-pass rate; 95% Wilson CI: 58.4%-82.0%), including 100% of failures (27/27) from fluent flagship/large generators under emergency pressure.
- **Eliminated Lexical False Alarms on Frozen Rules:** Resolved the 48.2% false-alarm rate (177/367) of untyped keyword matchers via a canonical Assertion/Negation Gate (PROPOSED vs. NEGATED), achieving 100.0% specificity (367/367) and 100.0% criterion sensitivity (Youden's J = 1.000) against pre-frozen guideline rule specifications.
- **Zero-Hallucination Extraction & Guideline Grounding:** Enforced character-level grounding (norm(span) in norm(R)) with 0 parser hallucinations across 1,767 extracted events across 50 emergency scenarios anchored to Class I AHA/ACC, ESC, BTS, EAST, and Surviving Sepsis guidelines.

2. Clinical Reasoning Platform & Domain-Independent Execution Kernel [Live Cloud Web App] | [Architecture Spec] | [Kernel Code]

Creator & Full-Stack Python Developer | Cloud Web Application (hammam16.pythonanywhere.com) & DAG Reasoning Engine (2026)

- **Cloud Clinical Reasoning Platform:** Built and deployed a live Python web application and modular runtime kernel that transforms unstructured clinical decision flows and chain-of-thought traces into Directed Acyclic Reasoning Graphs (DAGs) with explicit hypothesis-evidence provenance links.
- **Cognitive Bias & Structural Failure Detectors:** Implemented deterministic graph analyzers to localize epistemic failure patterns including anchoring bias, premature diagnostic closure, unsupported inference, and temporal prerequisite violations (idx(prereq) < idx(action)).
- **Contract-Driven Software Quality:** Enforced strict runtime boundary isolation, deterministic serialization round-trips, and zero-leakage contracts across 30+ automated pytest test suites.

PEER-CITABLE PREPRINT & TECHNICAL MEDICAL PUBLISHING

- **DOI Preprint (Zenodo / CERN):** Omer, H. (2026). Beyond Surface Fluency: A Deterministic-First Hybrid Clinical Auditor Exposes LLM-as-a-Judge Blindness and Behavioral Shifts Under Emergency Clinical Pressure. Official DOI: 10.5281/zenodo.23835945 (20 pages, 50 scenarios, 420 live evaluations).
- **Founder & Technical Author - NexoraMed (nexoramed.online):** Authored technical articles in English analyzing Medical AI evaluation methodologies, clinical reasoning frameworks, and AI tools in medical education (2026 - Present).

TECHNICAL & CLINICAL ARSENAL

- **AI Safety & LLM Evaluation:** Multi-Ecosystem Benchmarking, LLM-as-a-Judge Failure Auditing, Sycophancy & Precautionary Shift Analysis, Automated Rubric & Protocol Engineering, Statistical Verification (Wilson 95% CIs, Exact McNemar Tests, 6-Tier Ablation).
- **Software & Systems Engineering:** Python (OOP, Type-Safe Data Models, Cloud Web Apps, CLI & REST APIs), Neuro-Symbolic Pipelines, Deterministic Substring & Regex Engines, JSON Schema Constrained Decoding, Pytest TDD, Git/GitHub, Linux/Termux CLI.
- **Clinical Domain Grounding:** Emergency Medicine Triage & Resuscitation Protocols, Class I Guideline Interpretation (AHA/ACC, ESC, BTS, EAST, ATLS), Differential Diagnosis Modeling, Contraindication & Temporal Prerequisite Auditing.

EDUCATION

- **Bachelor of Medicine, Bachelor of Surgery (MBBS Candidate - 3rd Year):** Faculty of Medicine, Omdurman Islamic University - Omdurman, Sudan | Expected Graduation: 2029.